If You Give a Mouse a Cookie (Web)

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Description

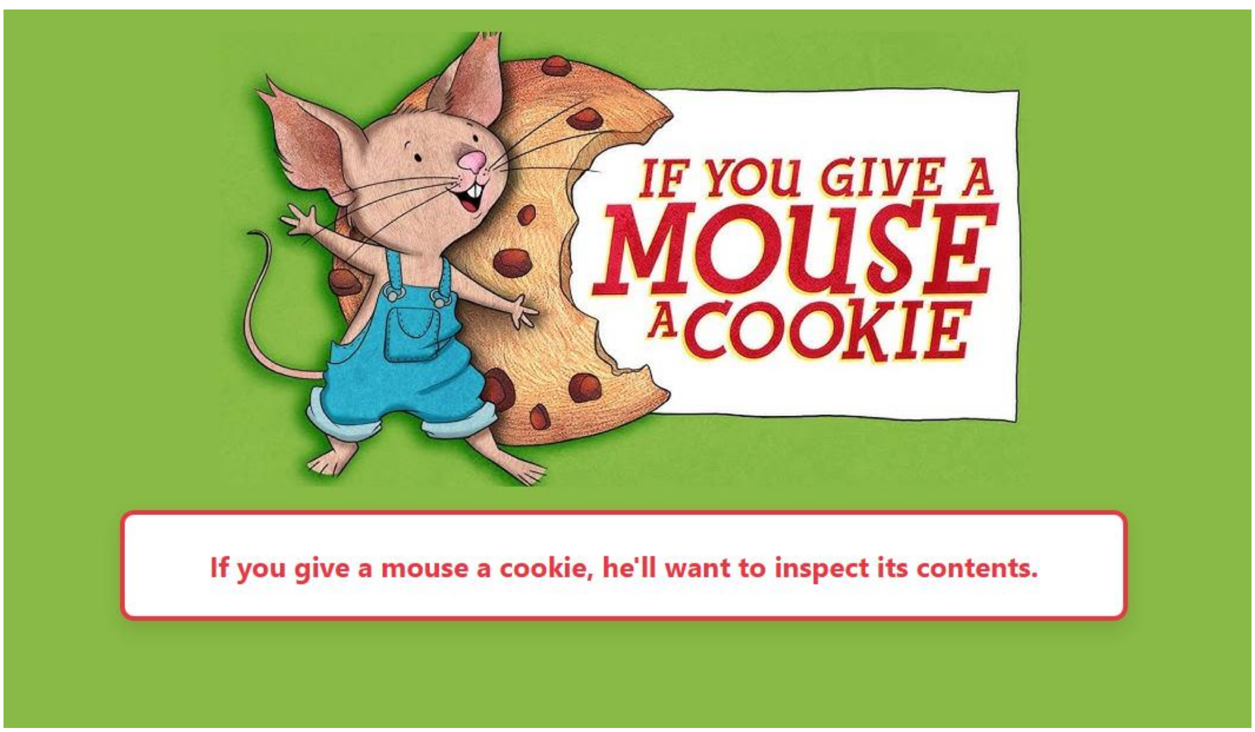
If you give a mouse a cookie, he might give you a flag in return.

Challenge Set-Up

Competitors are NOT given any challenge source code files. They are only given a link to the Docker instance run on the CTFd Infrastructure. The Docker instance can be set up via either Docker-Compose or the Dockerfile as specified in the CTFd challenge document. Both set-up options set up a Gunicorn WSGI server and EXPOSE port 8000 which can be edited or changed as necessary.

Writeup

The challenge begins with a webpage with the statement "If you give a mouse a cookie, he'll want to inspect its contents." This statement hints that there may be something valuable hidden within the cookie associated with the website.



Navigating to the cookie highlights that it has a base64 encoded value.

cookie

Name

cookie

Value

QWZ0ZXIgaW5zcGVjdGluZyB0aGUgY29udGVudHMsIGhlJ2xsIGhvcCBvb
iB0aGUgUk9CT1QgdmFjY3V1bSBwaWNraW5nIHVwIHRoZSBjcnVtYnMg
aGUgbWFkZS4KQ3J1bWIgMTogZFY5Q1FHc3paRjloVA==

Show Advanced

Placing it into a base64 decoder yields the below statement:

QWZ0ZXIgaW5zcGVjdGluZyB0aGUgY29udGVudHMsIGhlJ2xsIGhvcCBvbiB0aGUgUk9CT1QgdmFjY3V1bSBwaWNraW5nIHV
==

156 1

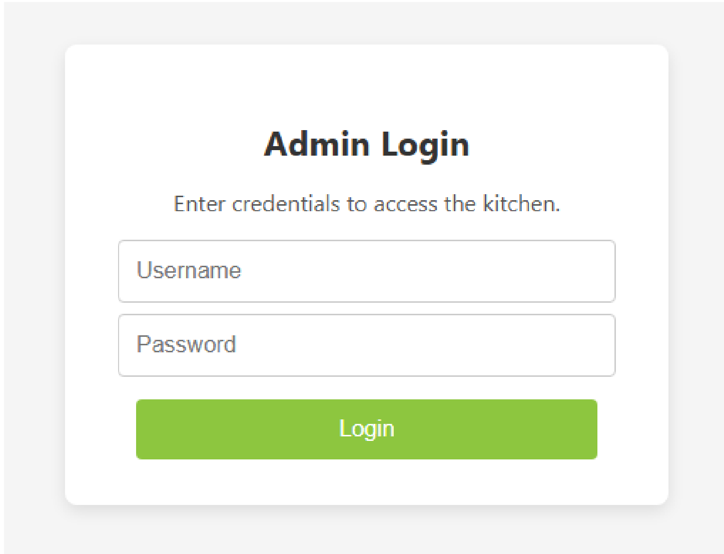
Output

```
After inspecting the contents, he'll hop on the ROBOT vaccuum picking up the crumbs he made.
Crumb 1: dV9CQGszZF9hT
```

The next clue, which mentions “ROBOTS” points towards the robots.txt path. Navigating here leads us to the next crumb, alongside a hint as to where to proceed. The hint in question is in reference to the /admin page, which is our hypothetical “locked door.”

```
User-agent: *
Disallow: /admin

# The robot vaccuum arrives at a locked door, which naturally he'll want to get inside
# Crumb 2: jB0SDNSX2MwMG
```

Navigating to /admin yields a login screen. When inspecting this page, located at the bottom of the source is a hardcoded script, which includes the username “admin,” and the password “sxM19mT3JfZEF”. Using these credentials successfully allows us to login.




```
<script>
    const ADMIN_USER = 'admin';
    const CRUMB_3 = 'sxM19mT3JfZEF';
    function login() {
        const user = document.getElementById('username').value;
        const pass = document.getElementById('password').value;
        if (user === ADMIN_USER && pass === CRUMB_3) {
            window.location.href = '/kitchen';
        } else {
            document.getElementById('error').style.display = 'block';
        }
    }
</script>
```

The next page we are brought to show a file directory. In /kitchen/ are a multitude of files, including a README.txt, which provides a hint for our next location, alongside stating that the password to the admin account is crumb 3.

Index of /kitchen/

Name	Last modified	Size
README.txt	2025-05-18 21:10	121
floor/	2025-05-18 21:10	-
pantry/	2025-05-18 21:10	-
refrigerator/	2025-05-18 21:10	-
stove/	2025-05-18 21:10	-

```
Breaking into the door will make him thirsty, he'll want to find a glass of MILK.

Crumb 3? Try the password on the door.
```

The hint tells us to locate a “glass of MILK.” Traversing through the directory leads us to the Milk.js file within /kitchen/refrigerator/. In this file there are three main items, another hint, some instructions, and crumb 4. The hint states that we will need a “cookie,” and the instructions tell us we must find a note to create one.

Index of /kitchen/refrigerator/

Name	Last modified	Size
../	-	-
700$_charcuterie_board.jpeg	2025-05-18 21:10	1652071
Milk.js	2025-05-18 21:10	316
rollCake.mp4	2025-05-18 21:10	116347302
veggies.mp3	2025-05-18 21:10	2217844

```
// If he asks for a glass of milk, he's going to want a cookie to go with it.

function pourMilk() {
    console.log("Pouring a fresh glass of milk... 🥛");
}

//TODO: Follow cookie instructions written on NOTE in /kitchen
function bakeCookie() {
    var crumb4 = "fTW9VNWUhISE=";

    return false;
}

pourMilk();
```

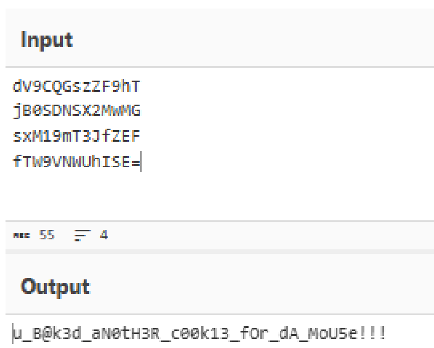
Heading back to the main kitchen directory, we can then navigate to /kitchen/floor/, which contains aNoteFallenFromTheFridge.txt. This note contains the recipe to create a cookie, which includes assembling all four crumbs, decoding it using Base64, and using the output to save to our own cookie within the home page for the flag.

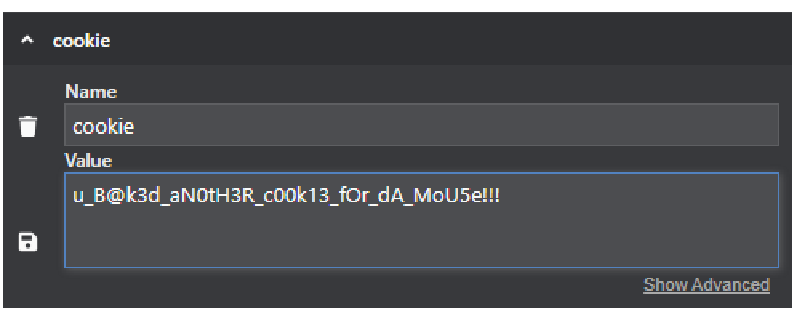
Index of /kitchen/floor/

Name	Last modified	Size
../	-	-
Dust (Another One).mp3	2025-05-18 21:10	5362426
aNoteFallenFromTheFridge.txt	2025-05-18 21:10	553
dust.jpg	2025-05-18 21:10	82084

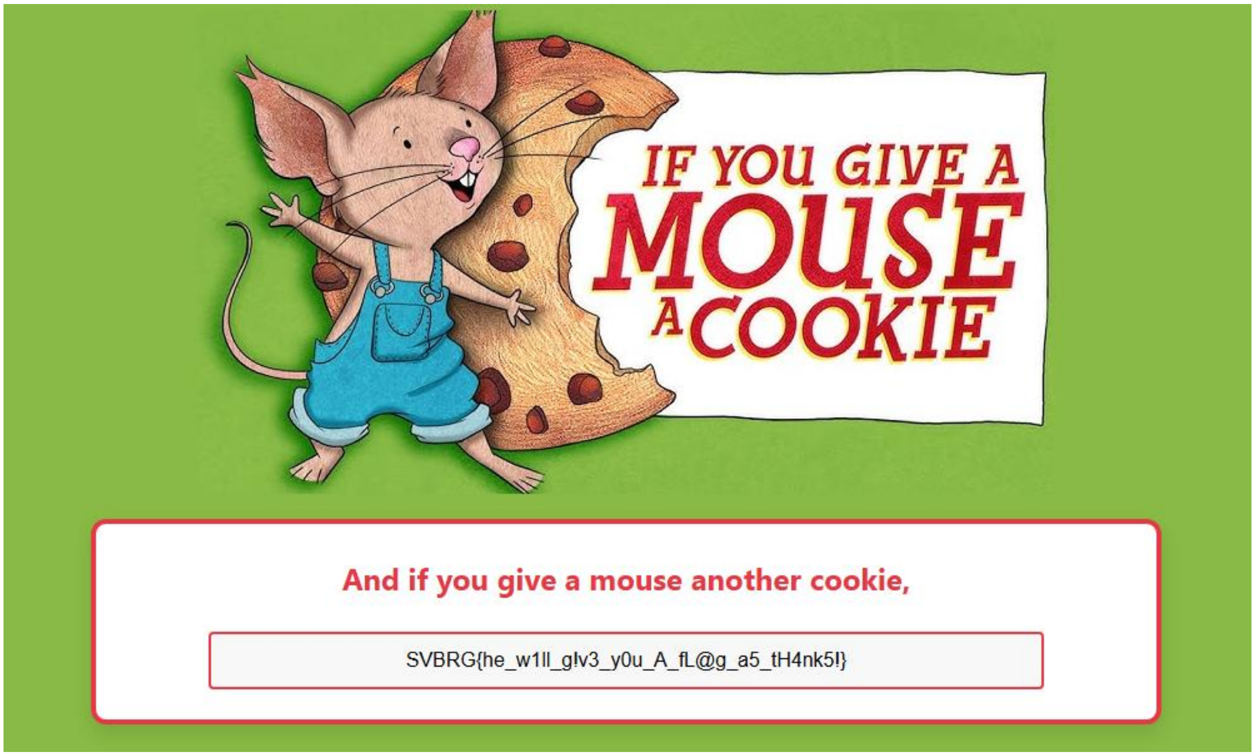
```
Steps to making my Fully Layered And Golden (FLAG) Cookie Recipe:

Ingredients:
- 4 Crumbs
- Butter
- Flour
- Sugar

Tools:
- Cookie Editor
- Large Mixing Bowl
- Sheet Pan
- Oven

Instructions:
1) Preheat oven to 320 degrees.
2) Assemble all 4 crumbs in order from 1 to 4.
3) Place assembled crumbs into the oven for 20 minutes to bake into a cookie.
4) Decode the assembled cookie using Base64 until nice and golden.
4) Use Cookie Editor to edit the original cookie with our new assembled cookie.
5) Refresh the home page.
6) Enjoy your FLAG cookie~!
```

Following these instructions, we can combine all four crumbs and decode, which yields the text "u_B@k3d_aN0tH3R_c00k13_fOr_dA_MoU5e!!!". From here, we can edit the cookie on the home page to insert this text into the value.





After the cookie has been saved, we can refresh the page to retrieve the flag.



FLAG: SVBRG{he_w1ll_g!v3_y0u_A_fL@g_a5_tH4nk5!}